A Mains-Derived Regulated Linear DC Power Supply: Design, Simulation and Bench Characterisation

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Report type	Technical report
Status	Designed, simulated, built and bench-tested
Tools	LTSpice, oscilloscope, electronic load
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Abstract. A 12 V, 1 A linear power supply was designed from first principles, simulated, built and tested. The supply comprises a step-down isolation transformer, a full-wave bridge rectifier, a reservoir capacitor and a three-terminal linear regulator. Closed-form relations were used to size the transformer, diode ratings, reservoir capacitance, regulator headroom and heatsink. For the chosen design point the calculated reservoir ripple is approximately 1 V peak-to-peak, the ripple trough of 14.6 V satisfies the regulator headroom requirement of 14.5 V, and regulator dissipation is approximately 3.1 W. LTSpice transient and load-step simulations were used to verify ripple and load regulation before construction. On the bench, ripple waveforms measured with an oscilloscope and output regulation measured with an electronic load were consistent with simulation; residual differences are attributed to parasitics omitted from the model.

Index terms: linear power supply, bridge rectifier, reservoir capacitor, ripple, linear regulator, thermal design, mains safety.

1. Objective and Specification

The objective was to size every stage of a linear supply from a quantitative requirement, verify the design in simulation, and compare the built circuit against the prediction. The specification is a regulated 12 V output at up to 1 A from 230 V, 50 Hz mains.

2. Architecture

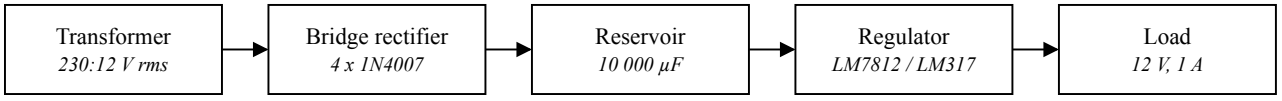


Figure 1. Four-stage linear supply. The transformer provides step-down and galvanic isolation; the bridge rectifies both half-cycles; the reservoir capacitor holds charge between mains peaks; the regulator rejects the residual ripple.

3. Sizing Relations

$$V_{pk} = \sqrt{2} \cdot V_{sec} - 2V_F \tag{1}$$

$$V_{ripple(pp)} \approx I_{load} / (2 f_{mains} C) \tag{2}$$

$$PIV \geq 2\sqrt{2} \cdot V_{sec}, \quad I_{FSM} \geq \text{inrush into a discharged capacitor} \tag{3}$$

$$V_{in,min} = V_{out} + V_{dropout} + \tfrac{1}{2} V_{ripple(pp)} \tag{4}$$

$$P_{reg} = (V_{in} - V_{out}) I_{out}, \quad \theta_{sa} \leq (T_{j,max} - T_a)/P_{reg} - \theta_{jc} - \theta_{cs} \tag{5}$$

Equation (4) applies at the ripple trough, where headroom is smallest. The transformer VA rating must cover the load power, losses and the RMS content of the narrow charging pulses, which exceeds the mean DC current [1].

4. Design Point

Table 1. 12 V, 1 A design point.

Stage	Value or part	Check	Evidence
Transformer	230:12 V rms, 15 VA	VA > 12 W load plus losses	Design choice
Rectifier	Full bridge, 1N4007	1000 V PIV >> 34 V; 30 A surge	Datasheet
Peak after bridge	15.6 V	Eq. (1)	Calculated
Reservoir	10 000 µF	Ripple 1 V_{pp}, Eq. (2)	Calculated
Regulator	LM7812 (fixed) or LM317 (adjustable)	Approx. 2 V dropout	Datasheet [3], [4]
Headroom	Trough 14.6 V	> 12 + 2 + 0.5 = 14.5 V	Calculated
Dissipation	3.1 W	(15.1 − 12) x 1 A, Eq. (5)	Calculated
Heatsink	Approx. 10 K/W	T_j well below 125 °C at 25 °C ambient	Calculated

The headroom margin is 0.1 V at full load, so the design is sensitive to low mains voltage; a larger reservoir or a higher secondary voltage would increase margin at the cost of dissipation.

5. Component Stress

The reservoir recharges only during a short interval near each mains peak, with conduction angle $\theta_c \approx \arccos(1 - V_{ripple(pp)}/V_{pk})$. The capacitor RMS ripple current is approximately $I_{C,rms} \approx I_{load}\sqrt{(\pi/\theta_c - 1)}$. For the design point, $\theta_c \approx 21°$ and $I_{C,rms} \approx 2.8$ A (calculated). Ripple-current rating, not capacitance alone, is therefore the selection criterion for the reservoir. Capacitor ESR adds a resistive ripple term and should be low.

6. Simulation

LTSpice transient analysis was used to quantify reservoir ripple against load current and to confirm that the regulator rejects it. A pulsed current source stepping between light and full load was used to assess load regulation. Iteration in simulation separated design errors from construction errors before hardware was built.

7. Bench Characterisation

The circuit was built and tested. Reservoir ripple was captured with an oscilloscope and compared with the LTSpice prediction; output voltage was checked across the intended load range using an electronic load. Measured ripple and load regulation were consistent with simulation. Residual differences are attributed to transformer winding resistance and leakage inductance, capacitor ESR and diode dynamic resistance, which the ideal model omits. Numerical measurement data are not included in this report.

Safety. Work referenced to the mains used transformer isolation, primary-side fusing, protective earthing of exposed metal, and single-hand probing with the supply isolated before changes [6].

8. Limitations

- Measured values are not tabulated against simulation, so agreement is stated qualitatively.
- Measurement uncertainty (probe attenuation, oscilloscope bandwidth) was not quantified.
- Line regulation and transient response to fast load steps were not characterised.

9. Conclusion and Future Work

The supply met its 12 V, 1 A specification in simulation and on the bench. The work illustrates the central trade-off of linear regulation: low noise and simplicity at the cost of efficiency, since the full input-to-output difference is dissipated. Future work: tabulate measured ripple and output voltage against load at several currents with stated uncertainty; measure line regulation; and evaluate a low-dropout regulator or a switching pre-regulator to reduce dissipation.

Appendix A. LTSpice Netlist

```
Vsec a 0 SIN(0 17 50)          ; 12 V rms secondary
D1 a p D
D2 0 p D
D3 n a D
D4 n 0 D
.model D D(Is=1n Rs=0.02)
Cres p n 10000u ic=15
Rbleed p n 10k
B1 p reg V=min(V(p)-2, 12)      ; behavioural regulator, 2 V dropout
Rout reg out 0.1
Iload out 0 PULSE(0.1 1 20m 1u 1u 40m 80m)
.tran 200m
.meas tran Vpp PP V(p)
.meas tran Vreg AVG V(out)
```

Listing A1. Netlist with a behavioural regulator; the bench build used an LM7812 or LM317.

References

[1] P. Horowitz and W. Hill, *The Art of Electronics*, 3rd ed. Cambridge University Press, 2015.

[2] A. S. Sedra and K. C. Smith, *Microelectronic Circuits*, 8th ed. Oxford University Press, 2020.

[3] Texas Instruments, "LM317 3-terminal adjustable regulator," datasheet SLVS044.

[4] Texas Instruments, "LM340, LM78xx series 3-terminal positive regulators," datasheet.

[5] M. H. Rashid, *Power Electronics: Circuits, Devices and Applications*, 4th ed. Pearson, 2013.

[6] IEC 61558, "Safety of transformers, reactors, power supply units and combinations thereof."